

AVIRAL MALIK

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PROFESSIONAL SUMMARY

Computer Science Engineering student with a strong foundation in data structures, algorithms, and system design, complemented by hands-on AI/ML experience across four internships building RAG pipelines, LLM applications, and edge AI systems. Proficient in Python, C++, FastAPI, and PyTorch. Seeking Software Engineering roles to apply core CS fundamentals and experience shipping production-grade, AI-integrated systems.

TECHNICAL SKILLS

Programming Languages: Python, Java, C/C++, MySQL, HTML

ML/AI Frameworks & Libraries: TensorFlow, PyTorch, LangChain, OpenCV, Hugging Face Transformers, scikit-learn, XGBoost, ChromaDB, Sentence-Transformers

Tools & Platforms: Streamlit, Flask, Gradio, Pandas, NumPy, Ollama, Verilator, ModelSim, SystemVerilog, Google Colab, MS Office, Postman, AWS, FastAPI, NestJS, Docker, MongoDB

Concepts: Data Structures, Algorithms, Object-Oriented Programming (OOP), Machine Learning, Deep Learning, Generative AI, Agentic AI, Computer Vision, Retrieval-Augmented Generation (RAG), Natural Language Processing (NLP), Fine-Tuning, Prompt Engineering, Reinforcement Learning, Reinforcement Learning from Human Feedback (RLHF), Edge AI

Languages: English, Hindi, Kannada, French (Beginner)

WORK EXPERIENCE

AI Engineer Intern

July 2026 – Present

XIBOTIX PRIVATE LIMITED | Tamil Nadu, India

- Developing edge AI and large language model applications.
- Assist in the optimization and deployment of AI systems for embedded platforms.
- Work on machine learning research, model optimization, and production AI pipelines.
- Utilized tools and languages including C++, Python, PyTorch, LLMs, Edge AI.

AI Agent Development Intern

May 2026 – July 2026

SkillWiz | Remote

- Architected a scalable AI grading engine using FastAPI, Gemini API, and Whisper to evaluate student submissions and calculate speech confidence metrics.
- Engineered bidirectional APIs linking a FastAPI engine with a NestJS backend to deploy autonomous AI agents, rigorously testing endpoints using Postman.
- Implemented strict LLM guardrails and Pydantic validation to ensure predictable AI outputs, containerizing the system for deployment on AWS EC2.
- Utilized tools and languages including Python, TypeScript, FastAPI, NestJS, MongoDB, Docker, AWS, and Postman.

AI & ML Intern

May 2025 – August 2025

Kineton Technologies | Tamil Nadu, India

- Developed C-based algorithms for image processing and compression, optimizing code for deployment on constrained hardware environments.
- Designed and simulated hardware accelerator modules using SystemVerilog to speed up machine learning inference tasks including decision tree execution.
- Explored emerging trends in AI-driven Electronic Design Automation (EDA) and automated hardware generation using Python and ModelSim.
- Utilized tools and languages including C, SystemVerilog, ModelSim, and Python.

AI & ML Intern

March 2025 – April 2025

Voxvertex | Tamil Nadu, India

- Developed a website-integrated chatbot using NLP and large language models (LLMs) to assist users with company information and website navigation.
- Built a semantic search prompt system enabling users to discover suitable guest lecturers for events using Hugging Face Transformers and BeautifulSoup.

- Developed a Generative AI tool to visualize event scenes from user descriptions using GANs and diffusion models (Diffusers library).
- Utilized Python, Flask, Hugging Face Transformers, BeautifulSoup, Diffusers, and Pandas.

PROJECTS

Reflexive RAG-RTL — AI-Enhanced RISC-V Hardware Generation System

April 2026

Tech Stack: Python, Qwen2.5-Coder (Ollama), ChromaDB, Verilog, Verilator, Sentence-Transformers

- Developed a Reflexive RAG pipeline integrating ChromaDB vector store and Verilator for automated linting, eliminating port-name hallucinations and reducing integration lint errors from 22 to 2 via a validation-gated feedback loop.
- Successfully synthesized a complete RISC-V core passing 38/38 ISA tests by implementing a multi-tier Trust Hierarchy to prioritize verified local module interfaces over LLM-generated assumptions.

Conflict-Aware Social Optimization Solver — Academic Research

March 2026

Tech Stack: Python, Branch and Bound, Lagrangian Relaxation, Quadratic 0-1 Knapsack

- Developed an exact Branch and Bound (B&B) solver using Lagrangian Relaxation to model multi-constraint guest selection as an NP-hard Quadratic 0-1 Knapsack Problem (QKP).
- Implemented a greedy warm-start and Lagrangian-guided pruning mechanism, evaluating less than 1.5% of the brute-force search space and achieving up to a 70x computational speedup.
- Recovered the provably optimal solution across all test configurations, completely eliminating severe negative conflict penalties produced by standard Dynamic Programming and Greedy baselines.

Vision Craft — AI-Based Real-Time Gesture Control System

December 2025

Tech Stack: Python, OpenCV, Google MediaPipe, NumPy, PyDirectInput

- Developed a hands-free gaming interface using Google MediaPipe to map specific hand gestures to keyboard and mouse inputs with DirectX compatibility via PyDirectInput.
- Engineered a custom virtual joystick algorithm with exponential smoothing to convert 2D hand-tracking coordinates into fluid 3D camera movement.

VidIntel — AI-Powered YouTube Video Intelligence Suite

June 2025

Tech Stack: Python, Streamlit, OpenAI Whisper, Hugging Face Transformers (Flan-T5, DistilBART), YouTube Data API, yt-dlp, GoogleTrans

- Developed a comprehensive multi-modal video analysis tool integrating automated audio transcription (OpenAI Whisper), text summarization, question answering, sentiment analysis, and translation.
- Built using Hugging Face Transformers (Flan-T5, DistilBART), YouTube Data API, yt-dlp, and GoogleTrans for multi-language content processing and deployed via Streamlit.

EDUCATION

B.Tech in Computer Science Engineering (Specialization in AI and ML)

2024 – Present

Vellore Institute of Technology, Vellore | CGPA: 9.73 | Ranked 3rd in Department of CSE-AIML

Grade 12 CBSE

2023 – 2024

National Public School, Bengaluru | 93.6%

Grade 10 CBSE

2021 – 2022

National Public School, Bengaluru | 90.6%

CERTIFICATIONS

- [Building Agents with Agentic AI](#) — IBM
- [Generative AI Engineering with LLMs](#) — IBM, Coursera
- [Deep Learning Specialization](#) — DeepLearning.AI, Coursera
- [Machine Learning Specialization](#) — Stanford University & DeepLearning.AI, Coursera
- [Python for Data Science, AI & Development](#) — IBM, Coursera